Project Report: Multimodal Perception & Reasoning System

1. Summary

This project presents a hybrid artificial intelligence framework designed to bridge the gap between high-speed object detection and complex semantic understanding. By integrating a discriminative perception layer (YOLOv8) with generative Vision-Language Models (Moondream and LLaVA), the system achieves real-time object tracking alongside a chronological reasoning narrative. The final output provides both a visually annotated video stream and a cohesive text-based intelligence summary of the entire recorded event.

2. System Architecture

The system is built upon a multi-stage pipeline designed to balance computational efficiency with analytical depth.

2.1 Perception Layer

- **Model:** YOLOv8n (Ultralytics)
- **Tracking:** ByteTrack
- **Functionality:** The perception layer handles the immediate task of object localization and identification. ByteTrack is utilized to ensure that temporal identities are maintained across frames, allowing the system to distinguish between unique subjects even in crowded scenes.

2.2 Reasoning Layer (Dual-Model Approach)

To optimize performance on local hardware, a hierarchical reasoning strategy was implemented:

- **Local Scene Narrative (Moondream):** Every 60 frames, the system invokes Moondream (a lightweight 1.6B parameter VLM). This model provides an immediate "play-by-play" description of the environment and subject actions, which is displayed in the live UI.
- **Global Historical Synthesis (LLaVA):** At the conclusion of the video stream, the accumulated narrative history is passed to LLaVA (a 7B parameter VLM). LLaVA acts as a final summarizer, reconciling chronological data to produce a single, professional summary report.

3. User Interface and Visualization

A critical component of the project is the custom-engineered side-panel interface developed using OpenCV.

- **Canvas Expansion:** The original video resolution is horizontally expanded by 400 pixels to accommodate a dedicated reasoning panel.
- **Live Narrative Display:** The panel displays the latest semantic output from the VLM, ensuring that the AI's reasoning is visible alongside the raw detection data.
- **Text Wrapping Engine:** A custom rendering function was developed to handle the multi-line generative text, preventing overflow and ensuring high legibility within the panel constraints.

4. Technical Implementation Details

The project was deployed locally on an Asus TUF Gaming F15 workstation, requiring specific optimizations for hardware efficiency:

- **Inference Gating:** By limiting VLM analysis to a periodic 60-frame interval, the system maintains a tracking speed of over 30 FPS while dedicating GPU resources to reasoning tasks only when necessary.
- **Metadata Harvesting:** Instead of storing high-resolution video segments for final analysis, the system harvests text-based metadata throughout the process. This significantly reduces memory overhead and simplifies the final synthesis stage.
- **Input/Output:** The system processes standard `.mp4` inputs and generates two distinct outputs: an annotated `.mp4` video with the side-panel UI and a `.txt` file containing the LLaVA-generated executive summary.

5. Case Studies and Semantic Observations

The reasoning layer proved effective in identifying semantic nuances that standard object detectors often miss.

- **Historical Context:** In footage containing historical attire, the reasoning layer was able to provide context beyond simple classification, identifying specific eras and attire functions.
- **Object Identification:** The system demonstrated a high degree of descriptive accuracy, identifying complex architectural features and decorative elements (e.g., distinguishing a large vessel as an "urn") that do not exist within the standard COCO dataset classes.

6. Conclusion and Future Directions

The Multimodal Perception and Reasoning System successfully demonstrates the viability of running advanced vision-language pipelines on localized hardware. By separating the perception and reasoning tasks into a hierarchical structure, the system provides both real-time utility and long-term analytical value.

Future iterations of this project will explore asynchronous multi-threading to eliminate the slight latency currently observed during VLM inference cycles and the implementation of specific "Safety Hazard" logic for industrial applications.